

Exercise 1: Treatment Design

You are planning an online experiment to test different design variations of your online store, seeking to learn which of them is most efficient to engage visitors on the site. As a first variable, you have come up with 4 different color schemes. As a second variable, you have created 3 different text introductions that greet the website's visitors.

- a) Sketch out a treatment matrix and discuss your major design choices for the experiment – including within/between subjects design, sample size, subject-to-treatment assignment, main outcome variable, etc.

Exercise 2: Dunning-Kruger Effect

You may have heard about the Dunning-Kruger effect (if not, make yourself familiar with it by a brief Internet research). In particular, you want to know whether telling people about this effect will reduce (or even increase it). In your own words, sketch out an experimental design to test whether this effect actually exists. Please note that you will have to recruit subjects from a University-based subject pool and will conduct the experiment at your institution's economics laboratory. Your research budget for paying off participants is 800€.

Be as specific as possible with regard to the following questions:

- a) What is the Dunning-Kruger effect?
- b) What are your main research questions?
- c) How does your treatment design look like?
- d) Which experimental task would you choose?
- e) How do you incentivize participants?
- f) What is the experiment's main outcome variable?
- g) How large should your sample be and how do you recruit participants?
- h) Which ethical considerations would you make?

Exercise 3: Peak-end rule

You may have heard of the *peak-end rule* (if not, please look it up). Your task now is to devise an experiment that tests this theory's claims in a clear-cut manner.

Hint: You will need to think of a specific task, a treatment design, a suitable sample, the precise procedure, stimulus materials, incentive structure, and measurement variables – among other things.

Exercise 4: Field Experiments

Given that field experiments offer many advantages, in particular regarding external validity, why should researchers not always conduct field experiments?